

**Table 1. SMOOTH CYLINDER — regime boundaries**

| Regime         | Re range                             | CD (approx)          | Strouhal St            | Wake character                                              |
|----------------|--------------------------------------|----------------------|------------------------|-------------------------------------------------------------|
| Sub-critical   | $Re < 3 \times 10^5$                 | $CD \approx 1.2$     | $St \approx 0.20$      | Laminar BL, turbulent wake, regular Kármán shedding         |
| Trans-critical | $3 \times 10^5 < Re < 3 \times 10^6$ | $CD \approx 0.3-0.9$ | Irregular              | Turbulent BL transition, "drag crisis", chaotic shedding    |
| Super-critical | $Re > 3 \times 10^6$                 | $CD \approx 0.6$     | $St \approx 0.25-0.35$ | Fully turbulent BL, narrower wake, organised shedding again |

**Table 2. 1.4 Force Components — Form vs Friction**

| Re range                  | Form drag fraction | Friction fraction | Dominant mechanism                 |
|---------------------------|--------------------|-------------------|------------------------------------|
| $Re < 1$                  | ~50%               | ~50%              | Stokes creeping flow               |
| $Re \sim 10^2$            | ~70%               | ~30%              | Attached flow, growing wake        |
| $Re \sim 10^4$            | ~90%               | ~10%              | Separated flow, large wake         |
| $Re \sim 10^5$ (sub-crit) | ~95%               | ~5%               | Turbulent wake, pressure dominates |

**Table 3. 2.2 Oscillatory Flow Regimes ( $Re = 1000$ , smooth cylinder — Sumer & Fredsøe)**

| KC range         | Regime name              | Flow description                                        | NL (lift freq./flow freq.) | Pairs per cycle |
|------------------|--------------------------|---------------------------------------------------------|----------------------------|-----------------|
| $KC < 1.1$       | No separation            | Purely viscous, fully attached flow                     | —                          | 0               |
| $1.1 < KC < 1.6$ | Honji instability        | 3-D Honji vortices along cylinder axis                  | —                          | 0               |
| $1.6 < KC < 4$   | Symmetric vortex pair    | Pair of symmetric attached vortices; no shedding        | —                          | 0               |
| $4 < KC < 7$     | Asymmetric attached pair | Vortex symmetry breaks; lift $\neq 0$ ; still attached  | 1 (approx)                 | 0               |
| $7 < KC < 15$    | Single-pair regime       | One vortex pair shed per full period; transverse street | 2                          | 1               |
| $15 < KC < 24$   | Double-pair regime       | Two vortex pairs shed per cycle, opposite sides         | 3                          | 2               |
| $24 < KC < 32$   | Three-pairs regime       | Three vortex pairs shed per cycle                       | 4                          | 3               |
| $32 < KC < 40$   | Four-pairs regime        | Four vortex pairs per cycle                             | 5                          | 4               |
| $KC > 40$        | Higher-pair regimes      | Each +8 KC adds one more pair per cycle                 | $n+1$                      | $n$             |

**Table 4. 2.3 NL Table Summary**

| KC regime | Pairs convecting away per cycle | NL = lift freq / flow freq | Alias        |
|-----------|---------------------------------|----------------------------|--------------|
| KC < 7    | 0 (attached or no separation)   | 1 (or irregular)           | Pre-shedding |
| 7 - 15    | 1 pair                          | 2                          | Single-pair  |
| 15 - 24   | 2 pairs                         | 3                          | Double-pair  |
| 24 - 32   | 3 pairs                         | 4                          | Three-pairs  |
| 32 - 40   | 4 pairs                         | 5                          | Four-pairs   |

**Table 5. 3.3 Force Components: Froude-Krylov vs Inertia vs Drag**

| Force component             | Formula                                               | Physical origin                             | When dominant                                        |
|-----------------------------|-------------------------------------------------------|---------------------------------------------|------------------------------------------------------|
| Froude-Krylov FK            | $\rho \cdot A \cdot (dU/dt)$                          | Undisturbed pressure gradient (hull absent) | Small structures ( $D/L < 0.2$ ), $CM \rightarrow 1$ |
| Hydrodynamic mass FHM       | $\rho \cdot C_m \cdot A \cdot (dU/dt)$                | Fluid inertia due to body presence          | Always present in water                              |
| Total inertia FI = FK + FHM | $\rho \cdot C_M \cdot A \cdot (dU/dt)$                | Both above combined                         | KC < 20; small cylinders                             |
| Drag FD                     | $\frac{1}{2} \cdot \rho \cdot C_D \cdot D \cdot U U $ | Pressure difference (wake) + friction       | KC > 20; large amplitudes                            |

**Table 6. 4.3 Cross-Flow VIV — Lock-in Thresholds (Steady Current)**

| Parameter               | Value                  | Condition / Notes                                                    |
|-------------------------|------------------------|----------------------------------------------------------------------|
| Onset of VIV            | $V_r \approx 4-5$      | Shedding frequency approaches $f_n$ ; vibration begins               |
| Lock-in range (air)     | $4.75 < V_r < 8$       | Shedding locks onto $f_n$ ; maximum amplitudes at $V_r \sim 5.5-6.5$ |
| Lock-in range (water)   | $3 < V_r < \sim 10$    | Broader range in water due to lower mass ratio effects               |
| Maximum A/D             | $\sim 0.5 - 1.5$       | Depends on $K_s$ ; peaks near $V_r \sim 5-7$                         |
| Unlock (end of lock-in) | $V_r \approx 7-8$      | $f_v$ jumps back to Strouhal frequency                               |
| Strouhal condition      | $f_v = St \cdot U_c/D$ | $St \approx 0.2$ in sub-critical; $V_{r\_lockin} \approx 1/St = 5$   |

**Table 7. 4.4 In-Line VIV — Instability Regions (Steady Current)**

| Region                  | Vr range                     | Cause                                                                                                                                          | Max Ax/D                             |
|-------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 1st instability         | $\sim 1.25 < V_r < \sim 2.5$ | Normal Kármán + symmetric secondary shedding;<br>$f_x = 3 \cdot St \cdot U_c/D \rightarrow f_x = f_n$ at<br>$V_r = 1/(3 \cdot St) \approx 1.7$ | $\sim 0.05-0.10$                     |
| 2nd instability         | $\sim 2.5 < V_r < \sim 3.8$  | Normal Kármán drag oscillation at $2 \cdot f_v$ ; $2 \cdot f_v = f_n$ at<br>$V_r = 1/(2 \cdot St) = 2.5$                                       | $\sim 0.10-0.15$                     |
| 3rd (cross-flow region) | $V_r \sim 5-8$               | Cross-flow VIV drives in-line at $2 \cdot f_y$ ; much larger Ax due to strong vortices                                                         | $\sim 0.3-0.5$ (when $A_y/D > 0.2$ ) |

**Table 8. Higher  $K_s \rightarrow$  more damping/heavier  $\rightarrow$  smaller VIV. Key design lever: add structural damping or mass.**

| $K_s$ value    | Expected cross-flow A/D | Design implication                                        |
|----------------|-------------------------|-----------------------------------------------------------|
| $K_s < 1$      | $A/D \rightarrow 1.0+$  | Large VIV; suppression devices needed (strakes, fairings) |
| $1 < K_s < 3$  | $A/D \sim 0.3-0.8$      | Moderate VIV; check fatigue life carefully                |
| $3 < K_s < 10$ | $A/D \sim 0.1-0.3$      | Small VIV; often acceptable                               |
| $K_s > 10$     | $A/D < 0.1$             | VIV suppressed; structure is very heavy or highly damped  |

**Table 9. Shallow water ( $h < L/20$ ):  $\tanh(kh) \rightarrow kh \rightarrow c = \sqrt{g \cdot h}$  (non-dispersive)**

| Quantity                  | Deep water formula                    | Shallow water formula                 | General                                          |
|---------------------------|---------------------------------------|---------------------------------------|--------------------------------------------------|
| Phase speed $c$           | $gT/(2\pi) \approx 1.56T$ [m/s]       | $\sqrt{g \cdot h}$                    | $\omega/k$                                       |
| Group velocity $c_g$      | $c/2$                                 | $c = \sqrt{g \cdot h}$                | $c \cdot n, n = \frac{1}{2}(1 + 2kh/\sinh(2kh))$ |
| Wavelength $L$            | $gT^2/(2\pi) \approx 1.56T^2$ [m]     | $T \cdot \sqrt{g \cdot h}$            | $2\pi/k$                                         |
| Max horiz. velocity $U_m$ | $\pi H/T = \omega a$ [at surface]     | $\pi H/(T \cdot kh) = c \cdot H/(2h)$ | $(\pi H/T) \cdot \cosh(k(z+h))/\sinh(kh)$        |
| Max vert. velocity $w_m$  | $\pi H/T$ (same as $U_m$ )            | 0 at bottom                           | $(\pi H/T) \cdot \sinh(k(z+h))/\sinh(kh)$        |
| Max horiz. accel.         | $\omega^2 a = (2\pi/T)^2 \cdot (H/2)$ | —                                     | $\omega^2 \cdot a \cdot \cosh(k(z+h))/\sinh(kh)$ |

**Table 10. 6.2 Integral Wave Parameters from Spectral Moments**

| Parameter                  | Symbol          | Formula                                                                | Physical meaning                               |
|----------------------------|-----------------|------------------------------------------------------------------------|------------------------------------------------|
| Significant wave height    | $H_s = H_{1/3}$ | $H_s = 4 \cdot \sqrt{m_0}$                                             | Mean of top 1/3 waves; "typical storm wave"    |
| Variance of surface        | $\sigma^2$      | $\sigma^2 = m_0$                                                       | Area under wave energy spectrum                |
| RMS surface elevation      | $\sigma$        | $\sigma = \sqrt{m_0}$                                                  | Standard deviation of $\eta(t)$                |
| Mean wave period ( $T_1$ ) | $T_1 = T_m$     | $T_1 = 2\pi \cdot m_0/m_1$                                             | Period at mean frequency                       |
| Mean energy period         | $T_E = T_{-1}$  | $T_E = 2\pi \cdot m_{-1}/m_0$                                          | Weighted by energy; used in power calculations |
| Zero-crossing period       | $T_z = T_2$     | $T_z = 2\pi \cdot \sqrt{(m_0/m_2)}$                                    | Mean time between zero up-crossings            |
| Mean crest period          | $T_c$           | $T_c = 2\pi \cdot \sqrt{(m_2/m_4)}$                                    | Mean time between wave crests                  |
| Peak period                | $T_p$           | $T_p = 2\pi/\omega_p$ ( $\omega_p = \operatorname{argmax} S(\omega)$ ) | Period at spectral peak                        |
| Significant period         | $T_s$           | $T_s \approx 0.95 \cdot T_p$ (Bretschneider)                           | Mean period of highest 1/3 waves               |

**Table 11. 6.3 Period Relationships for Standard Spectra**

| Spectrum type    | T <sub>z</sub> /T <sub>p</sub>                                    | T <sub>1</sub> /T <sub>p</sub> | T <sub>s</sub> /T <sub>p</sub> |
|------------------|-------------------------------------------------------------------|--------------------------------|--------------------------------|
| Bretschneider    | ≈ 0.71                                                            | ≈ 0.77                         | ≈ 0.95                         |
| JONSWAP (γ=3.3)  | ≈ 0.74                                                            | ≈ 0.78                         | ≈ 0.97                         |
| General relation | T <sub>z</sub> < T <sub>1</sub> < T <sub>E</sub> < T <sub>p</sub> | —                              | —                              |

**Table 12. Quadrant table for phase angle**

| Quadrant | Re sign | Im sign | φ formula                         | φ range     |
|----------|---------|---------|-----------------------------------|-------------|
| Q1       | + (>0)  | + (>0)  | φ = arctan(Im/Re)                 | 0° - 90°    |
| Q2       | - (<0)  | + (>0)  | φ = 180° - arc-<br>tan( Im / Re ) | 90° - 180°  |
| Q3       | - (<0)  | - (<0)  | φ = 180° + arc-<br>tan( Im / Re ) | 180° - 270° |
| Q4       | + (>0)  | - (<0)  | φ = 360° - arc-<br>tan( Im / Re ) | 270° - 360° |

# Table 13. Standard normal (z-table) key values

| $z = x/\sigma$ | $P(Z > z)$ [exceedance] | Practical meaning                                                                    |
|----------------|-------------------------|--------------------------------------------------------------------------------------|
| 1.00           | 15.87%                  | ~16% chance of exceedance per observation                                            |
| 1.28           | 10.00%                  | 10% exceedance                                                                       |
| 1.645          | 5.00%                   | 5% exceedance (1 in 20)                                                              |
| 2.00           | 2.28%                   | ~2% (close to 1 in 50)                                                               |
| 2.04           | 2.06%                   | $\sigma=\sqrt{1.5}$ , $a=2.5 \rightarrow z=2.04 \rightarrow \sim 2.1\%$ exam example |
| 2.33           | 1.00%                   | 1% exceedance (1 in 100)                                                             |
| 3.00           | 0.13%                   | ~1 in 800                                                                            |
| 3.72           | 0.01%                   | ~1 in 10,000                                                                         |

## Table 14. 7.1 Degrees of Freedom

| Motion | Direction        | Axis         | Restoring?                    |
|--------|------------------|--------------|-------------------------------|
| Surge  | Longitudinal (x) | —            | No (no hydrostatic restoring) |
| Sway   | Transverse (y)   | —            | No                            |
| Heave  | Vertical (z)     | —            | YES (waterplane area)         |
| Roll   | Rotation about x | Longitudinal | YES (transverse stability)    |
| Pitch  | Rotation about y | Transverse   | YES (longitudinal stability)  |
| Yaw    | Rotation about z | Vertical     | No                            |



**Table 15.  $GM_T = KB + BM_T - KG$** 

| Quantity           | Formula (rectangular box)     | Physical meaning                              |
|--------------------|-------------------------------|-----------------------------------------------|
| KB                 | $T/2$                         | Height of centre of buoyancy above keel       |
| $BM_T$             | $B^2/(12 \cdot T) = I_{wl}/V$ | Metacentric radius; $I_{wl} = L \cdot B^3/12$ |
| KG                 | Depends on load distribution  | Height of centre of gravity above keel        |
| $KM_T = KB + BM_T$ | $T/2 + B^2/(12T)$             | Height of metacentre above keel               |
| $GM_T$             | $KM_T - KG$                   | Transverse metacentric height                 |

**Table 16. 9.2 Scaling Laws (Froude Similarity, same fluid)**

| Quantity             | Scale factor    | Example: $\alpha=40$ |
|----------------------|-----------------|----------------------|
| Length L             | $\alpha$        | 40                   |
| Speed V              | $\sqrt{\alpha}$ | $\sqrt{40} = 6.32$   |
| Time T               | $\sqrt{\alpha}$ | 6.32                 |
| Force / Resistance F | $\alpha^3$      | 64,000               |
| Mass / Displacement  | $\alpha^3$      | 64,000               |
| Power P              | $\alpha^{3.5}$  | 10,119               |
| Wetted area          | $\alpha^2$      | 1,600                |

**Table 17. A. Dimensionless Numbers**

| Number             | Symbol | Formula                                        | Governs                         |
|--------------------|--------|------------------------------------------------|---------------------------------|
| Reynolds           | Re     | $U_c \cdot D / \nu$                            | Viscous vs inertia; flow regime |
| Strouhal           | St     | $f_v \cdot D / U_c \approx 0.2$                | Vortex shedding frequency       |
| Keulegan-Carpenter | KC     | $U_m \cdot T / D$                              | Separation in oscillatory flow  |
| Froude             | Fn     | $V / \sqrt{gL}$                                | Wave-making (ship)              |
| Reduced velocity   | Vr     | $U_c / (f_n \cdot D)$ or $U_m / (f_n \cdot D)$ | VIV lock-in proximity           |
| Mass ratio         | M      | $m / (\rho \pi D^2 / 4)$                       | VIV amplitude                   |
| Stability param.   | Ks     | $2 \cdot M \cdot 2\pi \cdot \zeta_s$           | VIV suppression                 |

**Table 18. B. Regime Thresholds — At a Glance**

| Regime type                           | Threshold                                  | Below →                                        | Above →                                               |
|---------------------------------------|--------------------------------------------|------------------------------------------------|-------------------------------------------------------|
| Re: Sub→Trans (smooth)                | $Re = 3 \times 10^5$                       | $CD \approx 1.2$ , $St = 0.2$ , regular Kármán | Drag crisis begins                                    |
| Re: Trans→Super (smooth)              | $Re = 3 \times 10^6$                       | Chaotic, no regular shedding                   | $CD \approx 0.6$ , $St \approx 0.3$ , organised again |
| KC: No sep → Honji                    | $KC = 1.1$                                 | No separation (creeping)                       | 3D Honji vortices                                     |
| KC: Honji → Symmetric pair            | $KC = 1.6$                                 | Honji instability                              | Symmetric attached pair                               |
| KC: Symmetric → Asymmetric            | $KC = 4$                                   | Attached symmetric vortices                    | Asymmetric; lift $\neq 0$                             |
| KC: Asymmetric → Single-pair SHEDDING | $KC = 7$                                   | Attached asymmetric vortices                   | VORTEX SHEDDING begins (NL=2)                         |
| KC: Single → Double pair              | $KC = 15$                                  | Single-pair (NL=2, 1 pair/cycle)               | Double-pair (NL=3, 2 pairs/cycle)                     |
| KC: Double → Triple pair              | $KC = 24$                                  | Double-pair (NL=3)                             | Three-pairs (NL=4, 3 pairs/cycle)                     |
| KC: Triple → Four pairs               | $KC = 32$                                  | Three-pairs (NL=4)                             | Four-pairs (NL=5)                                     |
| Each additional KC +8                 | Every 8 KC                                 | n pairs/cycle                                  | n+1 pairs/cycle                                       |
| VIV onset (cross-flow)                | $Vr \approx 4-5$                           | No significant VIV                             | VIV begins; approaches lock-in                        |
| VIV lock-in (cross-flow)              | $Vr \approx 5-7$                           | Pre-lock-in small amplitudes                   | Lock-in: $f_v = f_n$ , large A/D                      |
| VIV unlock                            | $Vr \approx 7-8$                           | Lock-in active                                 | $f_v$ jumps back to Strouhal                          |
| In-line: 1st instability              | $Vr \approx 1.25-2.5$                      | No in-line VIV                                 | 1st instability ( $A_x/D \sim 0.05-0.10$ )            |
| In-line: 2nd instability              | $Vr \approx 2.5-3.8$                       | 1st instability or no VIV                      | 2nd instability ( $A_x/D \sim 0.10-0.15$ )            |
| Wave breaking (Miche)                 | $H/L = 0.142 \cdot \tanh(kh)$              | Wave exists                                    | Wave breaks                                           |
| Diffraction important                 | $D/L = 0.2$                                | Morison equation adequate                      | Diffraction theory needed                             |
| Deep/shallow boundary                 | $h/L = 0.05$ (shallow); $h/L = 0.5$ (deep) | Shallow water: $c = \sqrt{gh}$                 | Deep water: $c = gT/(2\pi)$                           |

**Table 19. C. All Force Formulas**

| Force                   | Formula                                                   | Units | Notes                                 |
|-------------------------|-----------------------------------------------------------|-------|---------------------------------------|
| Drag (steady)           | $\frac{1}{2} \cdot \rho \cdot C_D \cdot D \cdot U_c^2$    | N/m   | Per unit length of cylinder           |
| Drag (oscillatory, max) | $\frac{1}{2} \cdot \rho \cdot C_D \cdot D \cdot U_m^2$    | N/m   | At time of max velocity               |
| Inertia (max)           | $\rho \cdot C_M \cdot (\pi D^2/4) \cdot U_m \cdot \omega$ | N/m   | At time of max acceleration           |
| Froude-Krylov (max)     | $\rho \cdot (\pi D^2/4) \cdot U_m \cdot \omega$           | N/m   | = FI,max / CM                         |
| FI/FD ratio             | $\pi^2 \cdot C_M / (K_C \cdot C_D)$                       | —     | Key exam shortcut                     |
| Buoyancy                | $\rho \cdot g \cdot V$                                    | N     | Upward force = displaced fluid weight |
| Righting moment (small) | $\Delta \cdot GM_T \cdot \sin(\varphi)$                   | N·m   | $\Delta$ = displacement weight        |